

SMART POWER DISTRIBUTION & CONSUMPTION ANALYTICS SYSTEM

Tutorial:- 3

TA:- Jamalpoor Sai Varun and Videh Sharma

Group :- 39

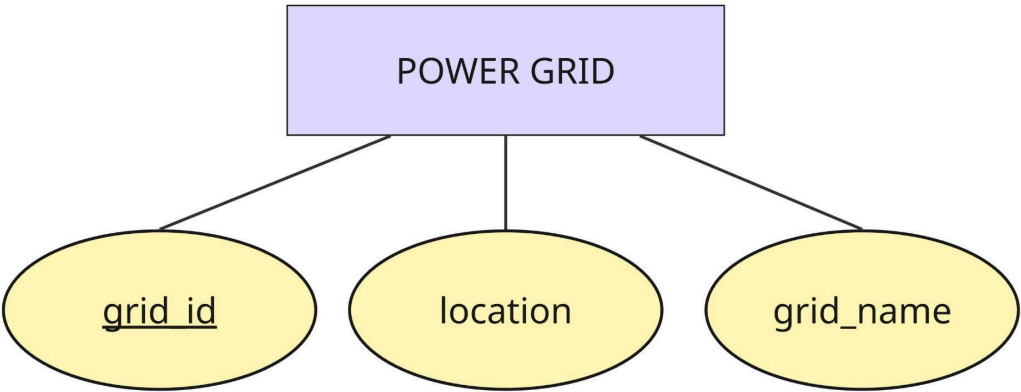
Task:-2

Team Members:-

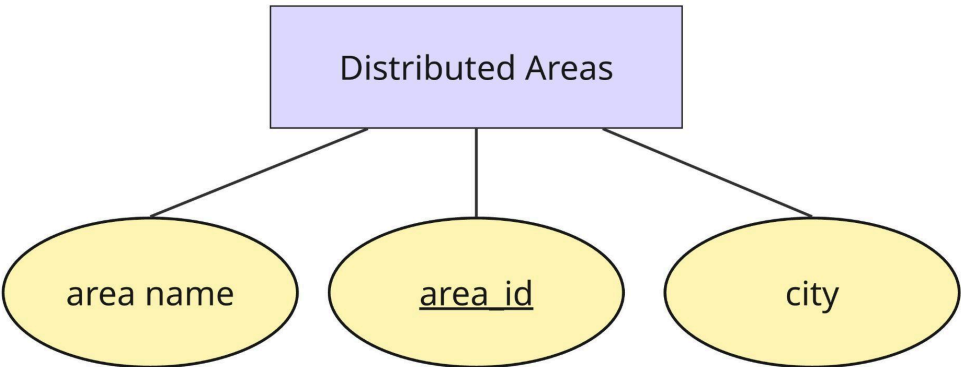
- Aditri Jain (2024027)
- Gaurav Jindal (2024221)
- Kushagra Agrawal (2024314)

Entities

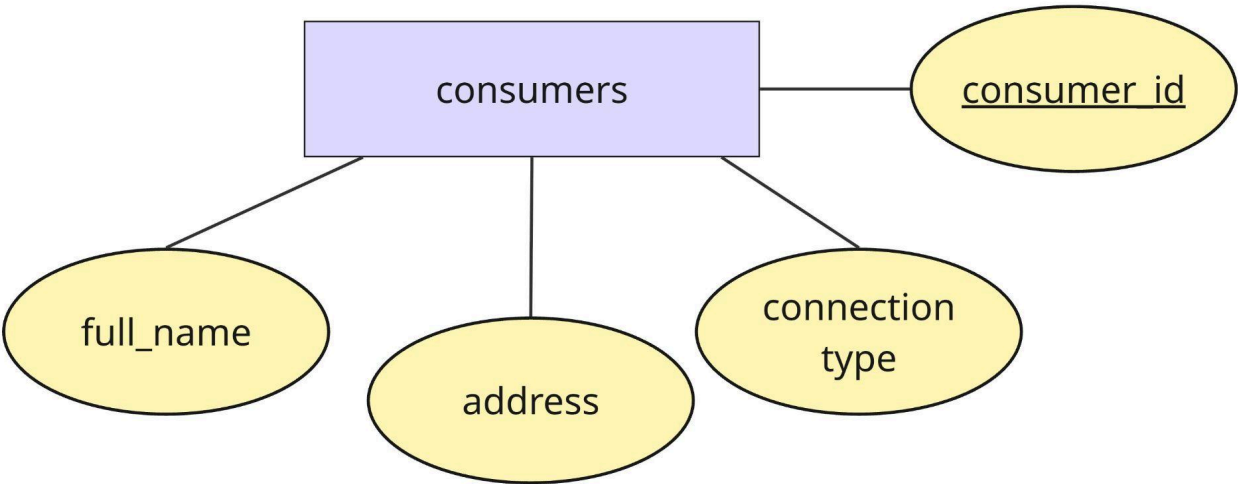
a)



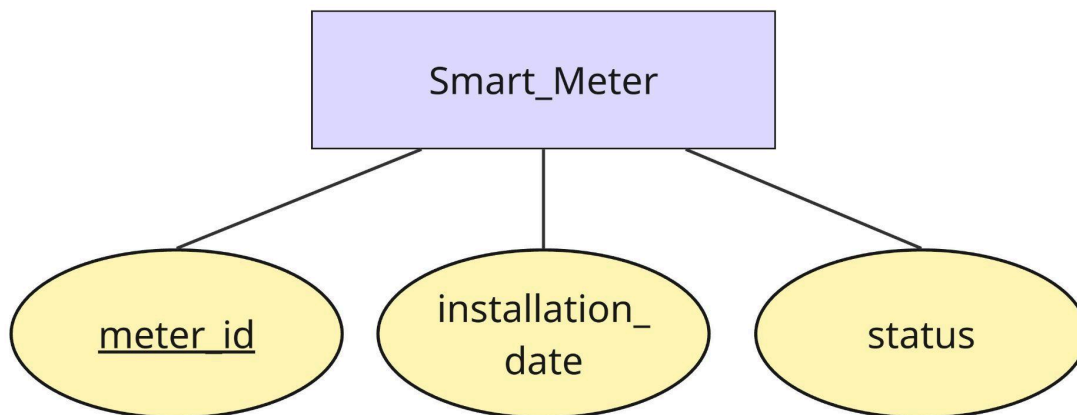
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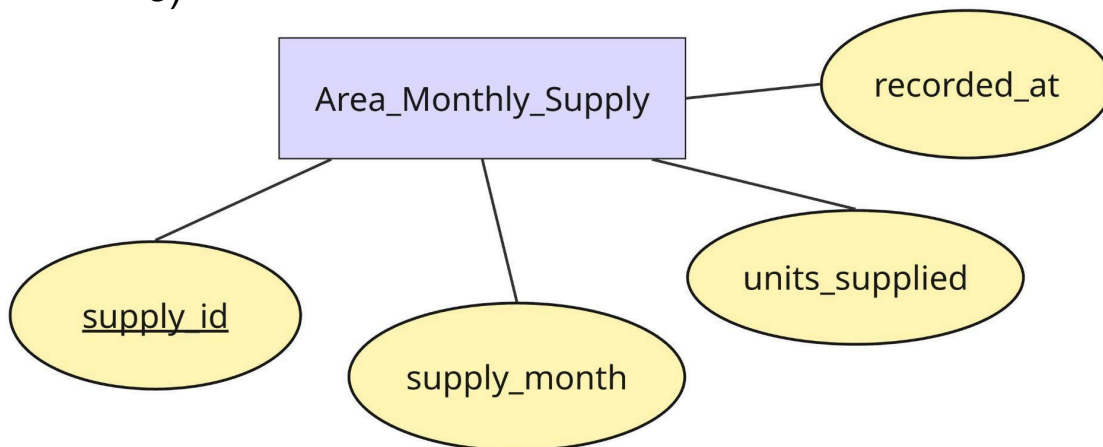
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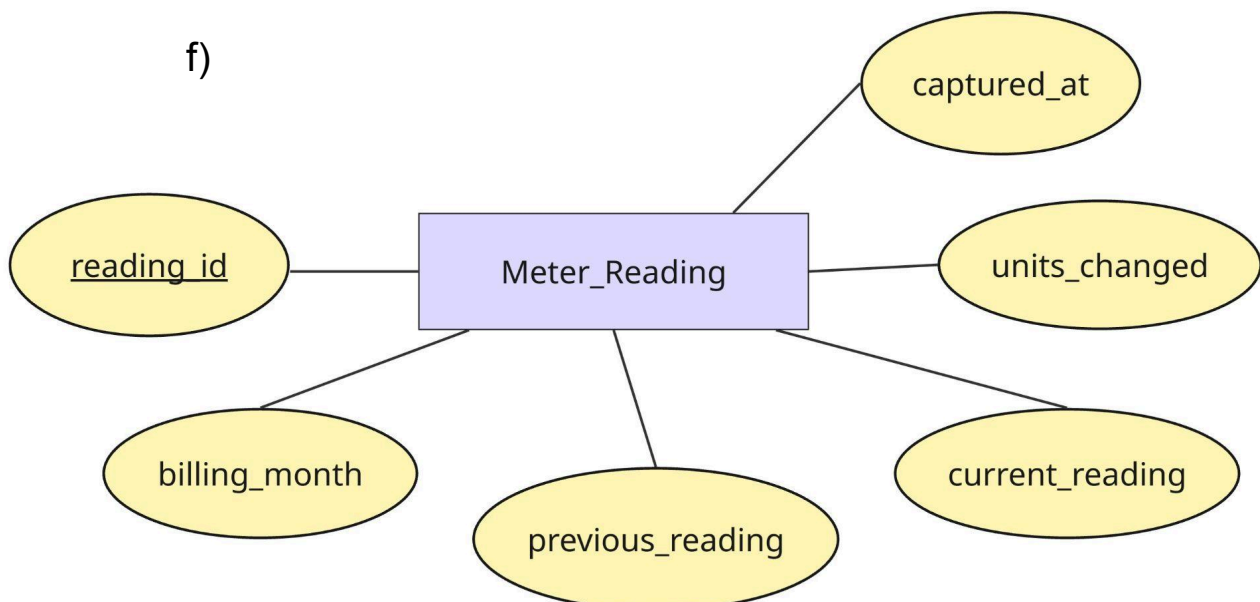
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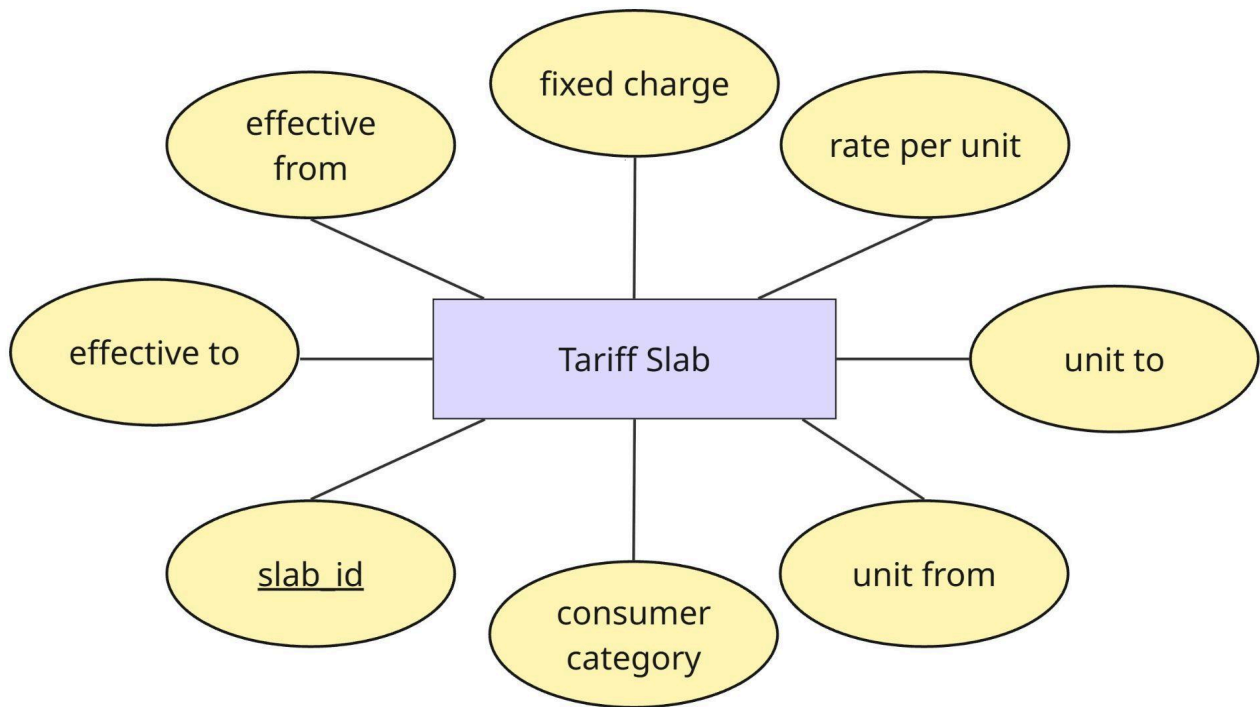
e)



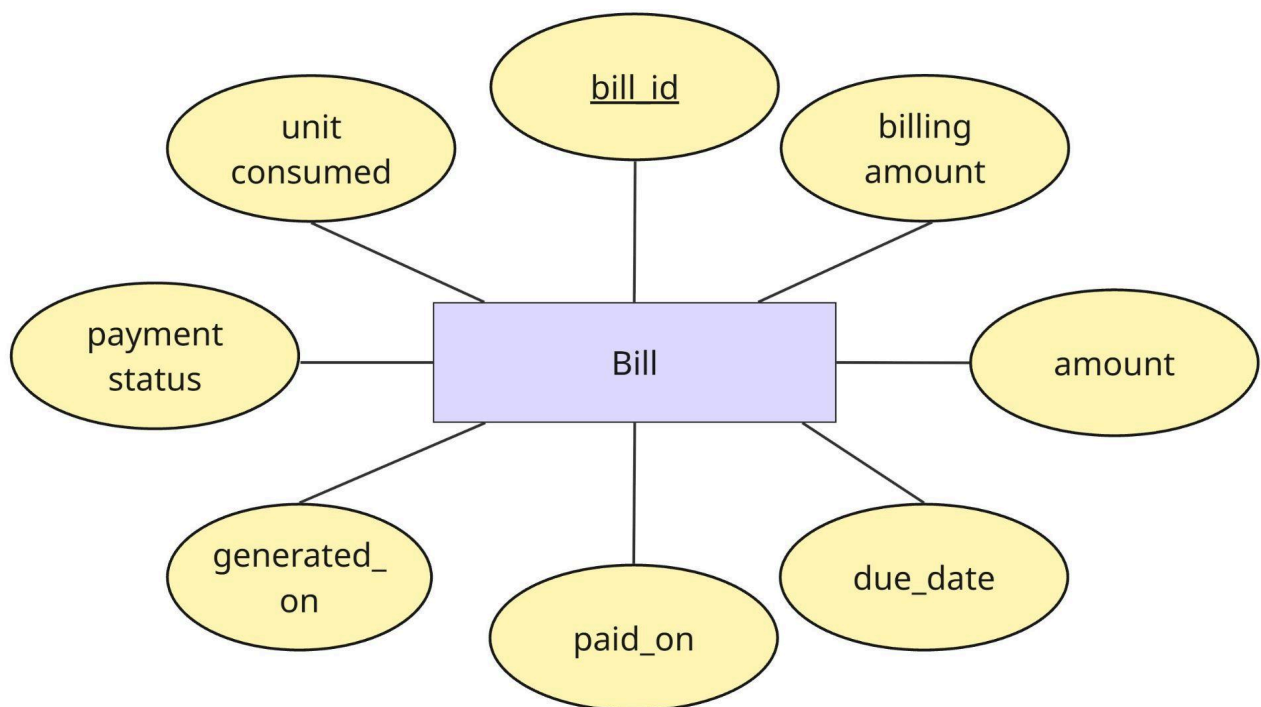
f)



g)

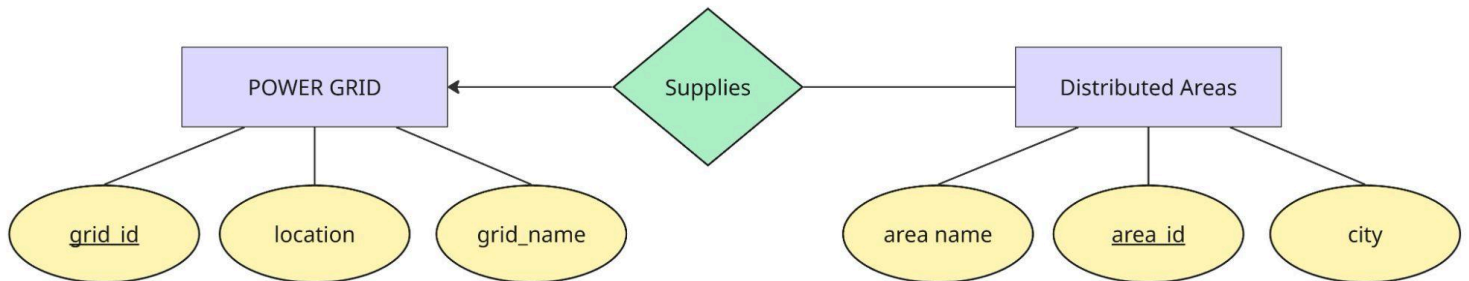


h)



Relationships

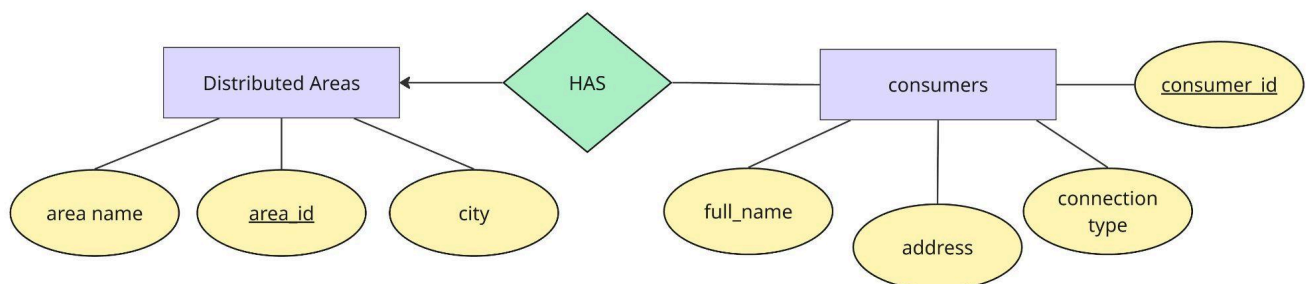
1) Power grid – (supplies) – distributed area



Relational Schema:-

DISTRIBUTED_AREA (
area_id PK,
grid_id Fk → POWER_GRID (grid_id),
area_name,
city
)

2)

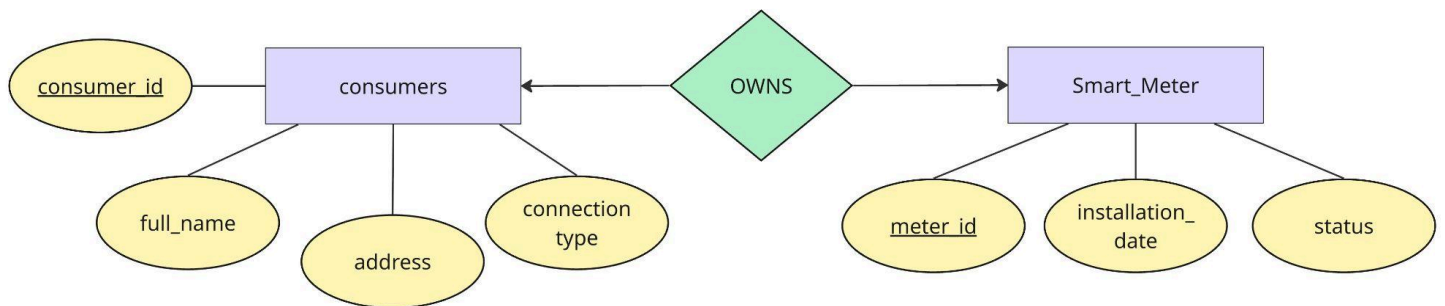


Relational Schema?

Consumer

consumer_id PK,
area_id FK → DISTRIBUTION-AREA (area_id),
full_name,
address,
connection_type
)

3)

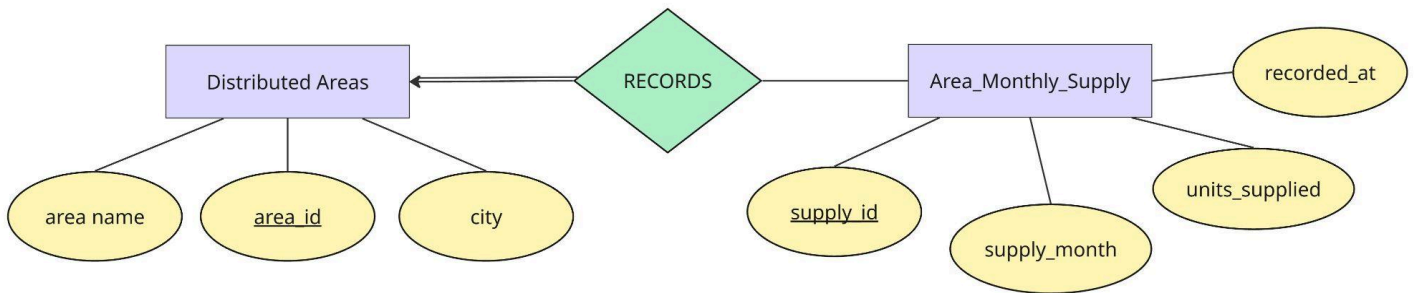


Relational Schema?

SMART METER (

meter_id PK,
consumer_id FK → consumer (consumer_id),
installation_date,
status,
)

4)



Relational Schema :-

AREA_MONTHLY_SUPPLY (

Supply_id pk,

Area_id fk → distribution area (Area_id)

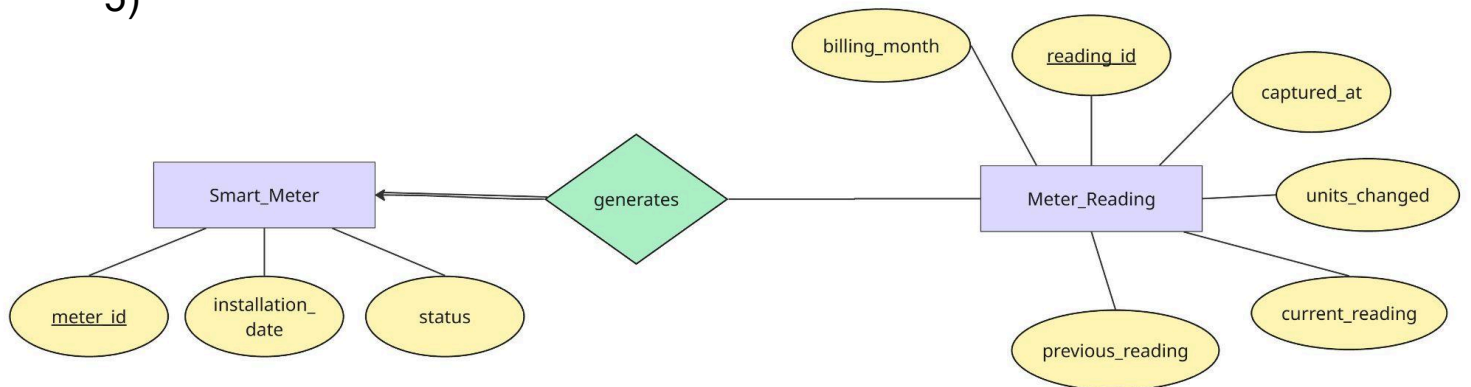
Supply_month,

units supplied,

recorded_at

)

5)



Relational Schema?

METER_READING (

reading_id pk,

meter fk, → SMART_METER (meter_id),

billing month,

previous reading,

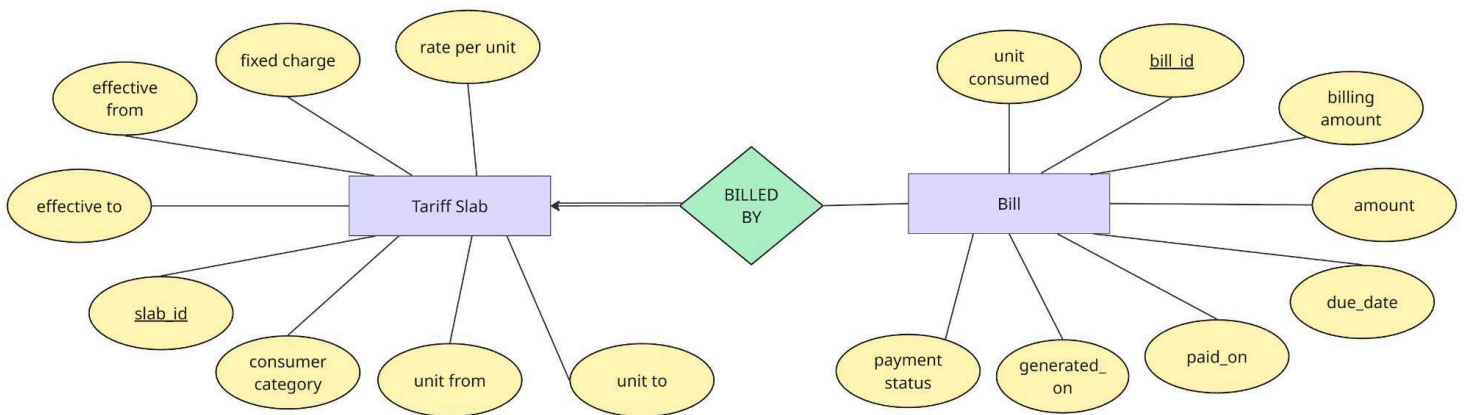
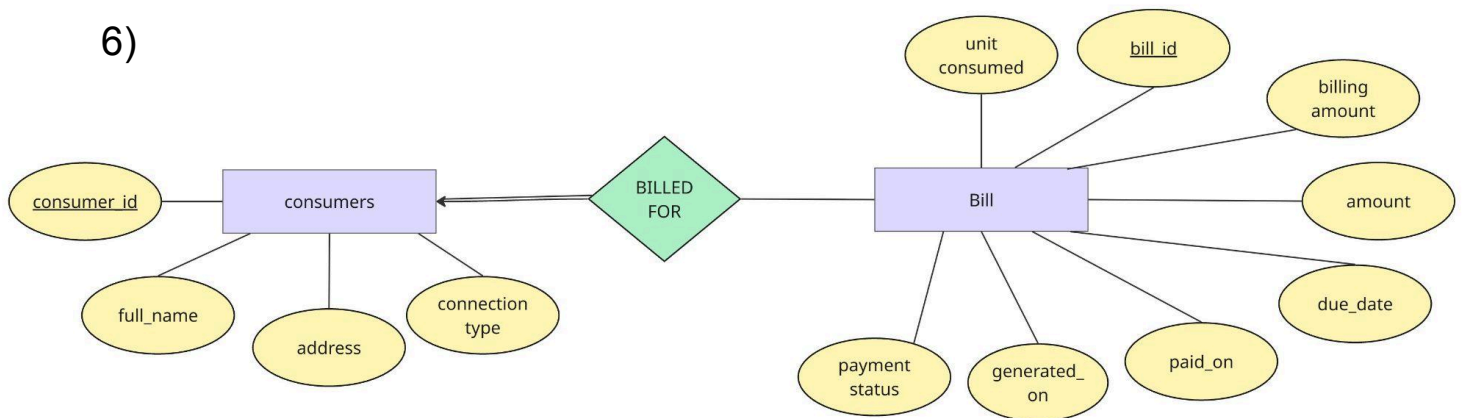
current reading,

units consumed,

captured at.

)

6)



Relational Schema?

o o o

BILL (

Bill id PK,

Consumer id PK → consumer(consumer id),

Slab id PK → TARIFF (slab id),

billing month,

unit consumed,

amount

payment status, generated on, due date, paid on)

Final ER Diagram

